

Solution Requirements

Date	1 July 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Secure login for authorized disaster management users.	High
2	Authorization levels	Admin, Meteorologist, Authority roles.	High
3	External interfaces	Weather dataset, Flask, IBM Cloud.	High
4	Transactions	Accept weather inputs and	High

	processing	generate flood prediction.	
5	Reporting	Prediction reports and flood risk dashboard.	High
6	Business rules, etc.	Predict flood risk using trained XGBoost model.	High
7	Compliance to laws or regulations	Protect user data and follow privacy standards.	Medium
8	<i>Other</i>	Store prediction history and model logs.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction in under 2 seconds.	<u>≤2 sec</u>
2	Scalability	Support multiple concurrent users.	1000+ users

3	Security & Data Privacy	Encrypted credentials and secure APIs.	HTTPS
4	Reliability & Availability	High availability of prediction service.	99.9%
5	Usability & Accessibility	Simple responsive UI.	Easy to use
6	<i>Other</i>	Maintainable and portable Flask app.	Modular